Apartment 50: Paper 010

AuthorityEffect: None. This is fictional examination staging, not a historical quotation, an examination attributed to Lev Landau, or a source of authority.

Landau asks for the ordered product. Woland changes the starting vertex. Behemoth presents a correct verdict without its certificate. The examiner asks what the host checked and which evidence supports the next claim.

Read the study guide, attempt the quiz, then consult the separate instructor key. Eight levels move from finite group arithmetic through lattice certificates to pilot reports. The capstone uses the package's host and retained evidence offline.

A receipt records its stated execution and custody checks. Its validity depends on the inputs, implementation and scope; a digest alone establishes no mathematical truth, physical claim or performance result. Give a calculation, a valid witness, a counterexample, or a precise missing obligation for each answer.

Study guide: Paper 010

AuthorityEffect: None. Study first, attempt the quiz, then use the separate instructor key. Definitions come from manuscript Section 2; pilot data and limits come from Sections 3–6.

1. Validate the algebra

A finite table must be total and closed, have a two-sided identity and inverses, and be associative for every triple. Names and values need their declared types. The S3 construction uses (p q)(x)=p(q(x)); Q8 uses integer quaternions. No floating-point tolerance enters these checks.

In Q8, i j (−i) (−j)=−1, while i(−i)j(−j)=1. The code labels 1 and −1 as `e` and `-e`. A −1 square is non-flat even though −1 is central and self-inverse.

2. Keep the path convention visible

Plaquettes are (r,c), (r+1,c), (r+1,c+1), (r,c+1). The last two edges use inverses of stored positive links. Reversal inverts the product. Moving the basepoint to the next vertex conjugates by the inverse of the first edge. Adjacent gauge factors cancel along a path, leaving `g(start) hol(path) inverse(g(end))`. A closed product is conjugated at its basepoint: identity status persists, but its exact element can change.

Use a non-central S3 product of order three to detect reversal errors. The Q8 commutator −1 alone cannot distinguish its two orientations.

3. Local flatness and global gauge triviality

The declared lattice includes its plaquettes. An open rectangle has trivial fundamental group, so flatness lets a rooted spanning-tree gauge trivialize every edge. On a torus, flatness leaves commuting cycle products A and B; both must be identity for a trivializing gauge. A=−1, B=1 therefore gives `FLAT_NONTRIVIAL`.

The host accepts open dimensions at least 2 and torus dimensions at least 3. Endpoint-only names cannot distinguish parallel torus edges at size 2. For seam links a,b, only the seam corner can be non-flat, with product b a b ^{1} a ^{1}. Enumerating $S3^{2}$ and $Q8^{2}$ covers 36+64=100 connections in this family, not all connections.

4. Optimization and proof boundary

Four identity links have identity product. Skipping their multiplications preserves the complete check; every other plaquette is still tested. The number skipped is gauge-dependent. Group validation takes $O(|G|^{3})$; after validation, the lattice pass and tree work take O(LM). A submitted path of k edges adds O(k) work.

Lean formalizes only the four-identity-link observation. The named report lists one axiom-free theorem. The historical full-module report lists 26 declarations: 25 axiom-free and one using `propext`, including generated structure declarations. Neither report formalizes the parser, flatness host or pilot analysis. Read the current receipt to distinguish static report checks from an executed build.

5. Require a witness

Verdict	Checked obligation	Near miss
NOT_FLAT	Actual plaquette with its matching nonidentity product	Correct label, wrong product or winding four-cycle
FLAT_TRIVIAL / PURE_GAUGE	Full gauge making every positive link identity	Missing vertex, conflicting alias or a remaining nonidentity edge
FLAT_NONTRIVIAL	All plaquettes flat and a closed path with matching nonidentity product	Open path, identity loop or non-flat input

The verifier accepts valid plaquette rotations and either orientation. Parsed edge or vertex aliases cannot overwrite entries. Unknown elements are refused before a verdict rather than treated as identity or silently omitted.

6. The right C017 denominator

Twenty-five historical responses are recorded. Fifteen solution records are retained: 13 parsed certificates and two parser-error wrappers. Fresh host checks accept 10 and reject 5 overall, including both wrappers. The other ten retain labels/reasons and zero recorded acceptances; they cannot be freshly certificate-checked. All 25 extracted entries can be compared with the original reports, which is a different check.

Arm 1's labels match 4/5 while certificates score 0/5 under the old harness. Three refusals concern vertex order before products were compared; those products are missing. The current host accepts either orientation but cannot recover absent payloads.

7. Audit historical tables

Table 3 reports original checker decisions out of ten per cell. C049 has two different panels; C050 repeats one panel. Table 4 uses the author's case-by-case classification (36 of 60 identifiers name a verdict/answer) and a prior checker

replay. Recounting does not independently assign these categories or measure model reliance on leakage. Readers can dispute the published annotations.

Always refusing passes 10/10 in original C045 and C046. Acceptance there cannot separate mathematical answers from that baseline. Paired checkers may share code or permissive failure modes; inspect obligations and negative controls before treating agreement as added assurance.

8. Name the execution supporting the claim

Operation	Evidence produced	Limit
C017 replay	Fresh verdicts and 15 solution-record checks	Ten payloads remain missing; no new responses
Pilot recount	Hashes, 25 extraction comparisons, Table 3/4 counts	Original flags and prior record; no fresh C045–C050 checker execution
Lean build/census	Result under the actual pinned compiler	Only the identity lemma; confirm the build ran
Deposit verification	Membership and byte custody	Does not establish mathematical truth
Hosted-model trial	New responses under recorded settings	Requires actual requests and suitable experimental design

Run the capstone and explain its fields. Future performance work needs opaque IDs, stripped answer notes, repaired checker refusal rules, negative controls, matched conditions/budgets, provider/error metadata, predeclared samples and complete retained responses. The package provides no new provider trial, learner study, physical result or operational authority.

Theoretical minimum: Paper 010

AuthorityEffect: None. Read the fictional preface and study guide. Keep the instructor key closed while answering. Use the manuscript's vertex order and group convention. In the Q8 code, `e` means 1 and `-e` means −1.

Each of 24 questions earns 3 points: result, calculation/evidence, and condition/limit. The capstone adds 8 points, for 80 total. These are practice criteria; no learner outcomes were measured.

Level 1: Group data before a verdict

1. A Q8 square has ordered links i, j, −i, −j. Compute its product and give a permutation of these factors that changes the product.
2. Why check associativity before using a supplied table? How many triples occur for S3 and Q8? State the asymptotic validation cost.
3. A table has all the right row labels, but one product is a list rather than an element name. May the host issue a verdict? Name the refusal type.

Level 2: Orientation, basepoint and gauge

4. Derive the effects of reversing a closed path and rotating its basepoint from v0 to v1. Why does question 1's Q8 example hide some orientation mistakes?
5. The canonical witness of C017 `pois_01` has product `r2`. Its reversed witness states `r`. Must the package accept it? What additional checks are required?
6. Show cancellation in a gauge-transformed path product. Which aspects of a closed holonomy persist, and which can change?

Level 3: Local and global flatness

7. Every plaquette of a valid open 2×3 lattice is flat. Explain why a spanning-tree gauge trivializes every edge. State the host's minimum open dimensions.
8. A flat Q8 torus has A=−1 and B=1. Give its verdict and explain why local flatness does not supply a globally trivial gauge.
9. A 3×4 Q8 torus has column seams a=i, row seams b=j and identity links elsewhere. Compute b a b ^{1} a ^{1} at the seam corner. Is it flat?

Level 4: Optimization and formal scope

10. A trivial 4×5 torus has 20 skippable plaquettes. After a gauge differing at neighbouring vertices, none is skippable. What changes and what persists?
11. Separate group-table validation cost from lattice checking. How does verifying a supplied loop depend on its length?

12. What does the named Lean lemma establish? Distinguish its one-declaration census from the historical full-module census of 26 declarations and from the Python host.

Level 5: Certificates and malformed inputs

13. A reply states `FLAT_TRIVIAL` without a gauge. Its label matches the host. Should the certificate verifier accept it? State the obligation.
14. A mapping contains `0,0->0,1` and `00,0->0,1`; a gauge contains `0,0` and `00,0`. Why must later entries not overwrite earlier ones? What should happen to unknown or non-string link values?
15. Give a positive certificate and a near miss for each verdict. Why is an arbitrary four-cycle insufficient as a `NOT_FLAT` plaquette witness?

Level 6: Reading C017

16. Five runs of five responses are recorded. How many solution records can be checked afresh? Separate parsed certificates from parser-error wrappers. Reconcile accepted/rejected counts with ten missing payloads.
17. Arm 1 has four labels equal to the host and zero accepted certificates. What do its orientation refusals establish? What cannot be reconstructed?
18. One torus response is unfinished at 8,000 tokens; another arm's response is accepted at 2,810. What conclusion fits this case? What confounds prevent a general causal claim about token savings?

Level 7: Reading the formal pilots

19. Distinguish C049 panels A/B from C050 runs 1/2. Why do these rows not establish a ranking of the three models?
20. Recount the six author-annotated identifier rows: how many of 60 name a verdict or answer? Does this show that models relied on the names?
21. A constant refusal passes 10/10 under both original C045 and C046 checkers. What does this imply for interpreting acceptance and the second checker?

Level 8: Reproduction and evidence boundaries

22. Distinguish fresh C017 host replay, Table 3 recount, Table 4 prior replay record, a fresh Lean build and a new hosted-model trial.
23. A ZIP matches its manifest. Which obligations are established, and which still require checks? How should an unexecuted pinned toolchain be recorded?
24. Propose a pilot assessing unaided performance and a prompt change. Address leakage, checker defects, matched budgets, provider metadata, sampling and retained artifacts. Do not infer that the proposed pilot has been run.

Capstone: An offline evidence packet

From the package root:

```
python tools/study_capstone.py --output /tmp/paper010-capstone.json
python code/replay_c017.py
python code/recount_pilots.py
```

On Windows, choose a writable output path. Inspect the JSON and write an eight-point audit: Q8 order; S3 reversal; seam verdicts; gauge-dependent skipping; certificate controls; malformed-input refusals; denominators and historical Table 3/4 counts; and the exact execution scope. A green exit alone is not an answer.

Instructor key: Paper 010

AuthorityEffect: None. Use after the quiz. A plausible label alone earns at most one of each question's three points.

Levels 1–2

1. i j (−i) (−j)=k(−i)(−j)=(−j)(−j)=−1. Reordering to i(−i)j(−j) gives 1.
2. Associativity supports group/path reasoning and is not implied by totality. S3 has 6^3=216 triples, Q8 has 8^3=512; validation costs O(|G|³).
3. `GroupTableError`: a list is not a declared element and may be unhashable. No verdict.
4. Reversal gives hol ^{1}; rotation gives U01 ^{1} hol U01. Central self-inverse −1 stays unchanged under either operation.
5. Accept if it is an actual plaquette, its listed-order product is `r`, and it is nonidentity. The old harness accepted only one convention.
6. Adjacent g(v) ^{1}g(v) cancel, leaving endpoint factors. Identity status and conjugacy class persist for a closed path; the exact element may change.

Levels 3–4

7. Set g(root)=e and g(v)=g(u)Uuv along the tree. Flatness permits plaquette path substitutions, so path independence on the rectangle extends this relation to all edges. Each open dimension must be at least 2.
8. `FLAT_NONTRIVIAL`: nonidentity A cannot be conjugated to identity.
9. j i (−j)(−i)=−1, so the corner and connection are `NOT_FLAT`.
10. Skipping changes 20→0; flatness, gauge triviality and verdict persist.
11. Validation O(|G|³); lattice/tree/gauge work O(LM); a k-edge certificate path O(k).
12. Four identity links give identity. The named theorem has no axioms; the full historical report has 26 declarations, 25 axiom-free and one `propext`. The complete checker and empirical analysis are outside the proof.

Levels 5–6

13. Reject: `gauge missing`. A matching label does not meet the witness obligation.
14. These spellings parse to one edge/vertex. Reject duplicates, even equal ones. Unknown/non-string links receive `LatticeError` before inverse lookup.
15. Accept an actual non-flat plaquette/product; reject a wrong product. Accept a full trivializing gauge; reject a missing vertex. Accept a flat torus with a matching closed nonidentity loop; reject an open or identity loop. A four-cycle can wind on a torus without bounding a plaquette.
16. 25 recorded responses; 15 retained solution records, 10 accepted and 5 rejected afresh. Thirteen records are parsed certificates, two are parser-

error wrappers. Ten missing payloads retain historical failures; their fresh checks are not run.

17. The messages establish convention-based rejection before product comparison, not a wrong product. Absent payloads cannot be reconstructed.
18. This shorter prompt produced an accepted response within its limit on this case. Several prompt changes, budgets, provider/sampling conditions and one reply per arm prevent a general savings or causal estimate.

Levels 7–8

19. C049 changes cases (stealth 5→9), so it is not a matched repeat. C050 repeats ten cases (7→6); five outcomes change, two involving API errors. Tasks, small authored panels, leakage and absent statistical design preclude model ranking.
20. 7+5+4+10+5+5=36/60. This is the author's identifier classification, not a measured dependence of model replies on leaked answers.
21. Model counts do not exceed the refusal baseline. The second checker accepts vacuous/unsupported outputs in the cited controls; agreement alone supplies no general independent assurance.
22. C017 replay executes algebra/certificate checks. Table 3 recount reads old flags; Table 4 recount reads annotations/prior checker outcomes. A fresh Lean build invokes the pin. A new trial obtains new provider responses.
23. Membership and byte identity are checked. Correctness, proof scope, design and rendering need separate evidence. Record actual failed/not-run toolchain status and retain earlier successful receipts as historical.
24. Use held-out cases with opaque IDs and stripped notes; repair/check refusal rules and both checkers; match conditions/budgets; record providers, errors and repeated samples; predeclare comparisons and retain complete outputs. This proposes work rather than reporting its execution.

Capstone rubric: eight points

One point for each audit item named in the quiz, supported by a calculation, output field or source. S3 has 18 commuting seam pairs and Q8 has 40: 58 flat of 100, comprising 2 `FLAT_TRIVIAL`, 56 `FLAT_NONTRIVIAL`, and 42 `NOT_FLAT`. Gauge skipping is 20→0. C017 counts stay 15/10/5 with ten missing payloads. Historical table recounts preserve Tables 3/4 and do not execute the formal checkers. The capstone runs neither Lean nor a provider. An 80-point practice score or receipt grants no further authority.